



Understanding PID Control

Introduction to Proportional, Integral, and Derivative Terms and Their Roles in Control Systems:

PID (Proportional-Integral-Derivative) control is a widely used control system in drones to maintain stability and achieve precise movement. Each term in the PID controller plays a distinct role in ensuring the drone responds correctly to control inputs and environmental disturbances.

Proportional (P) Term

The proportional term generates an output that is directly proportional to the current error. The error is the difference between the desired setpoint (target value) and the current value (measured by sensors). The proportional term responds to the magnitude of the error by applying a correction that is scaled by a proportional gain factor (K_p). The larger the error, the larger the proportional response.

Formula:

$$P = K_p \times \text{Error}$$

Example: Imagine the drone is flying at an altitude of 10 meters, but the desired altitude (setpoint) is 15 meters. The error is 5 meters (15 - 10). The proportional term calculates the necessary correction:

$$P = K_p \times 5$$

If the proportional gain (K_p) is set to 2, the output correction would be:

$$P = 2 \times 5 = 10$$

This correction is applied to the motor speeds to lift the drone higher. The closer the drone gets to the setpoint, the smaller the error becomes, and thus the smaller the correction.



Integral (I) Term

The integral term addresses accumulated errors over time. It sums up past errors to correct any residual error that the proportional term alone cannot eliminate. This is particularly useful for correcting steady-state errors, where the drone might consistently hover slightly below or above the setpoint due to persistent disturbances or biases.

Formula:

$$I = K_i \times \sum \text{Error}$$

Example: Consider a scenario where the drone consistently drifts 1 meter below the desired altitude due to a slight imbalance. Over time, this error accumulates:

$$\text{Total Error} = \sum \text{Error} = 1 + 1 + 1 + \dots$$

The integral term calculates the necessary correction to eliminate this steady-state error. If the integral gain (K_i) is set to 0.5, and the accumulated error over 10 seconds is 10 meters:

$$I = 0.5 \times 10 = 5$$

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This additional correction is applied to the motor speeds to bring the drone back to the correct altitude.

Derivative (D) Term

The derivative term predicts future error based on its rate of change. It provides a damping effect that helps reduce overshoot and oscillations by responding to the speed of error change. This term is particularly effective in stabilising the drone's movements by anticipating and counteracting rapid changes in the error.



Formula:

$$D = K_d \times \frac{d(\text{Error})}{dt}$$

Example: Imagine the drone is ascending too quickly towards the desired altitude. The derivative term calculates the rate of change of the error. If the error decreases rapidly, the derivative term produces a corrective action to slow down the ascent and prevent overshooting.

If the derivative gain (K_d) is set to 1, and the rate of change of error ($d(\text{Error})/dt$) is 3 meters per second:

$$D = 1 \times 3 = 3$$

This correction is applied to decrease the motor speeds, ensuring the drone approaches the setpoint more smoothly without oscillating around it.

Combining P, I, and D Terms

Full PID Controller Formula:

$$\text{Output} = K_p \times \text{Error} + K_i \times \sum \text{Error} + K_d \times \frac{d(\text{Error})}{dt}$$

By combining the proportional, integral, and derivative terms, the PID controller provides a comprehensive control strategy that addresses both immediate and long-term errors while predicting and dampening future errors. This combination ensures that the drone maintains stable flight and responds accurately to control inputs.

Example: Let's integrate all three terms for a comprehensive control action. Suppose the drone is at an altitude of 12 meters, the desired altitude is 15 meters, and the errors over the past few seconds have accumulated to 10 meters with a rate of change of -2 meters per second:



- **Proportional Term:** $P = 2 \times (15 - 12) = 2 \times 3 = 6$
- **Integral Term:** $I = 0.5 \times 10 = 5$
- **Derivative Term:** $D = 1 \times -2 = -2$

The combined output to the motors would be:

$$\text{Output} = 6 + 5 - 2 = 9$$

This output increases the motor speed to lift the drone towards the desired altitude, corrects accumulated errors, and stabilises the ascent to avoid overshooting.